Machine Learning-Based Website Fingerprinting Attacks on Tor Network

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Project Goals:

- To investigate whether websites can be identified in the Tor network
- To evaluate machine learning models for traffic classification
- To compare performance across different datasets

Research Question

Is it possible to identify websites on the Tor network using traffic features?

Sub-questions:

- Which model performs best?
- How dataset size affects performance?

Datasets overview

Dataset 1 — CSV

- 16 websites
- 5 sessions per site
- Small dataset
- Feature-based format

Dataset 2 — NPZ

- Large dataset
- Based on Tor Traffic Fingerprinting research
- More complex structure

Why two datasets?

- To compare performance on small vs large data

Data Processing

Data Preprocessing Steps:

- Feature extraction
- Label encoding
- Train/Test split
- Scaling (for SVM)

Machine Learning Models

Models Used:

- Random Forest
- Support Vector Machine (SVM)
- Convolutional Neural Network (CNN)

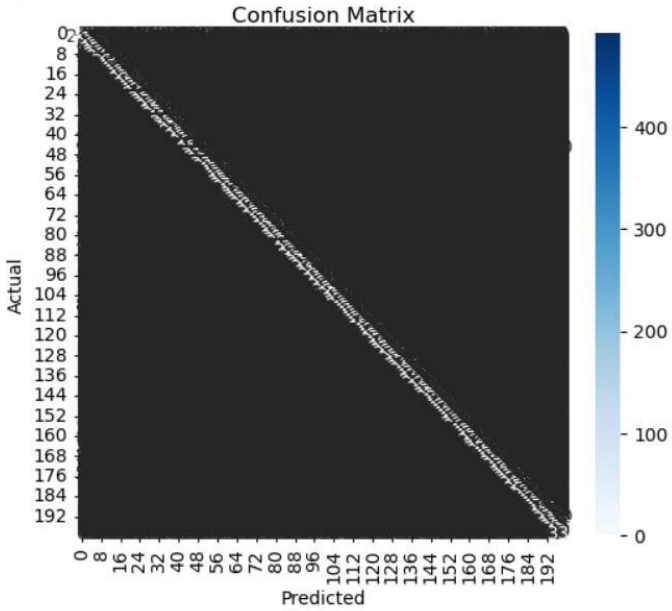
Evaluation Metrics

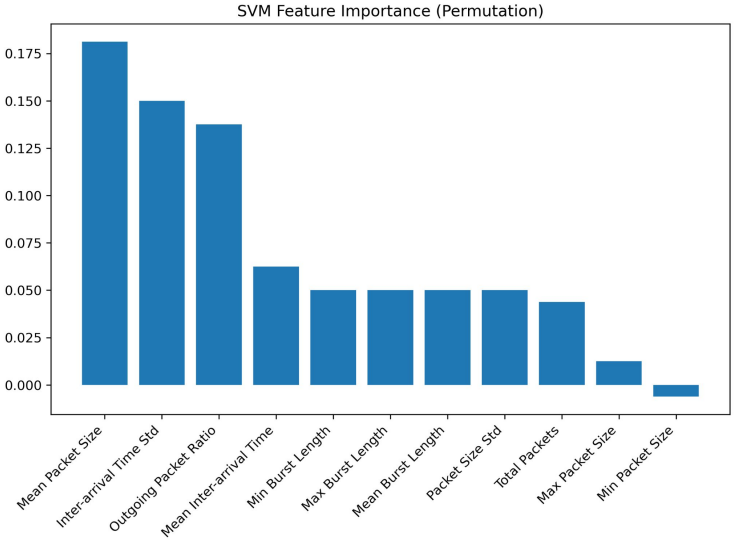
- Accuracy
- Precision
- Recall
- F1-score
- ROC Curve
- Confusion Matrix

Results Obtained

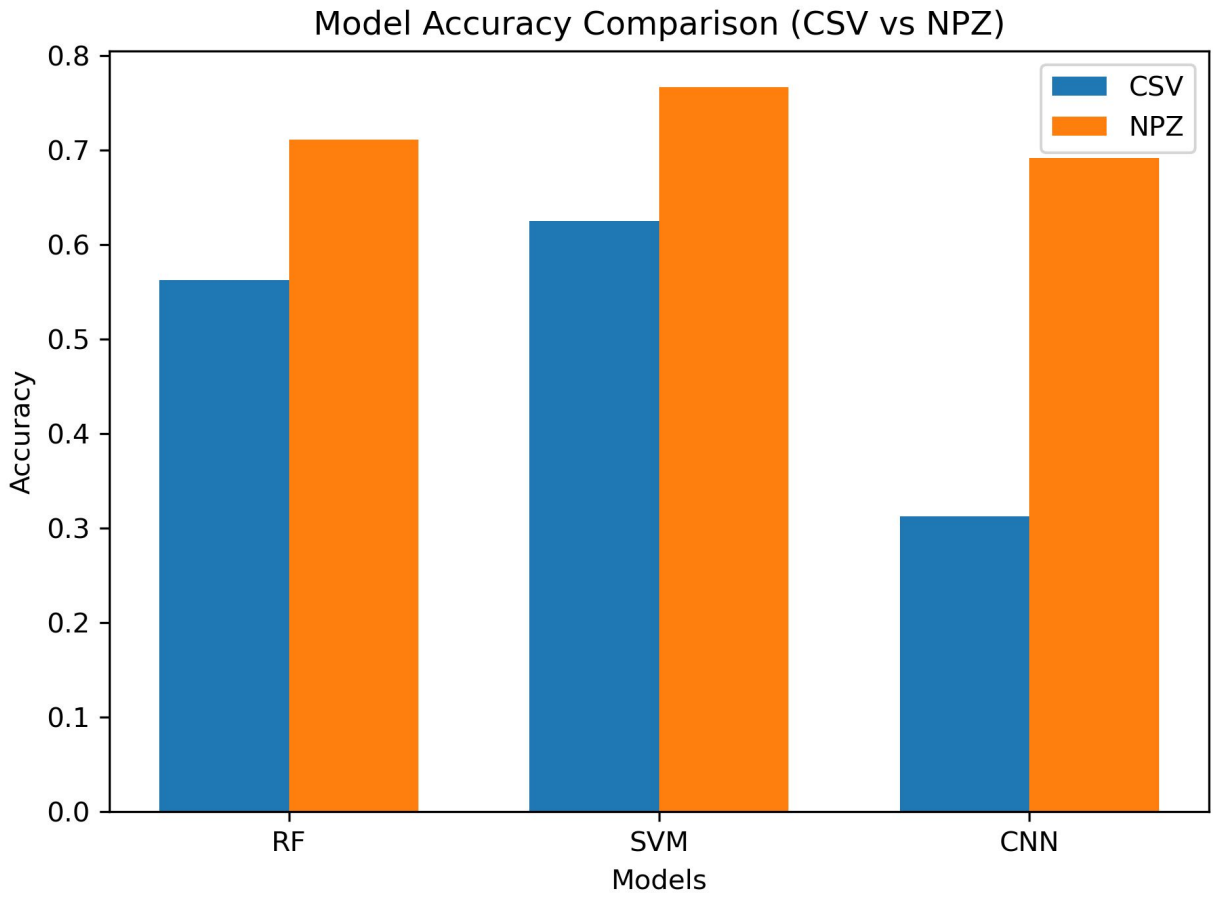
		Accuracy	**Precision**	**Recall**	**F1-score**
Model	**Datatype**				
RF	CSV	0.5625	0.4688	0.5625	0.5
	NPZ	0.7111	0.7303	0.7111	0.7018
SVM	CSV	0.625	0.4688	0.5625	0.5
	NPZ	0.7668	0.7974	0.7668	0.7716
CNN	CSV	0.3125	0.3489	0.5	0.3896
	NPZ	0.6921	0.702	0.6921	0.6819

Visual Results





Confusion matrix for SVM (NPZ)


Model Accuracy Comparison (CSV vs NPZ)
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
Accuracy
CSV
NPZ
RF
SVM
CNN
Models

Current Status

- **Completed:**
  - Dataset preparation
  - Model implementation
  - Model training
  - Performance evaluation
  - Result visualization
- **Next Step:**
  - Documentation
  - Result analysis
  - Final presentation preparation
- **Result: Currently ahead of schedule by ~2 weeks**